TOPSTECHNOLOGIES
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PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 2
Introduction to Programming
SESSION / ASSIGNMENT – 6

TASKS :

1. **Write a program that prints the numbers from 10 down to 1 using a for loop, similar to a countdown timer.**

➔

```c
#include <stdio.h>

int main()
{
    int i;

    for(i = 10; i >= 1; i--)
    {
        printf("%d\n", i);
    }

    return 0;
}
```

OUTPUT:

```
10
9
8
7
6
5
4
3
2
1
```

2. **Create a menu-driven console app that lets the user: 1) View your favorite 3 IPL teams, 2) Add a new team, 3) Exit. Use a while loop to keep showing the menu until the user chooses Exit.**

   Hint: Use input() (or Scanner in Java) to get the user's choice each time.

➔

```
#include <stdio.h>
#include <string.h>

int main()
{
    char teams[4][30] = {"Mumbai Indians", "Chennai Super Kings", "Royal Challengers
Bengaluru"};
    int choice;
    int count = 3;

    while(1)
    {
        printf("\n--- IPL Team Menu ---\n");
        printf("1. View Favorite Teams\n");
        printf("2. Add New Team\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        if(choice == 1)
        {
            printf("\nFavorite IPL Teams:\n");
            for(int i = 0; i < count; i++)
            {
                printf("%d. %s\n", i + 1, teams[i]);
            }
        }
        else if(choice == 2)
        {
            if(count < 4)
            {
                printf("Enter Team Name: ");
                scanf(" %[^\n]", teams[count]);
                count++;
                printf("Team Added Successfully!\n");
            }
            else
            {
                printf("Maximum team limit reached.\n");
            }
        }
        else if(choice == 3)
```

```
        {
            printf("Exiting...\n");
            break;
        }
        else
        {
            printf("Invalid Choice!\n");
        }
    }

    return 0;
}
```

OUTPUT:

```
--- IPL Team Menu ---
1. View Favorite Teams
2. Add New Team
3. Exit
Enter your choice:
```

```
Enter your choice: 1

Favorite IPL Teams:
1. Mumbai Indians
2. Chennai Super Kings
3. Royal Challengers Bengaluru
```

```
Enter your choice: 2
Enter Team Name: Gujrat Titans
Team Added Successfully!
```

```
Favorite IPL Teams:
1. Mumbai Indians
2. Chennai Super Kings
3. Royal Challengers Bengaluru
4. Gujrat Titans
```

```
Enter your choice: 3
Exiting...
```

3. **Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it. Use a do-while loop so the user can keep guessing until they get it right.**

   Constraint: Use at least 3 song names of your choice.

➔

```c
#include <stdio.h>
#include <string.h>

int main()
{
    char songs[3][30] = {"Believer", "Perfect", "Shape of You"};
    char guess[30];

    do
    {
        printf("\nGuess the Song: ");
        scanf(" %[^\n]", guess);

        if (strcmp(guess, songs[0]) != 0)
        {
            printf("Wrong Guess! Try Again.\n");
        }

    } while (strcmp(guess, songs[0]) != 0);

    printf("Correct! The song was %s.\n ", songs[0]);

    return 0;
}
```

OUTPUT:

```
Guess the Song: Poison
Wrong Guess! Try Again.

Guess the Song: Believer
Correct! The song was Believer.
```

4. **Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.**

➔ **1. entry-controlled loop (for/while)**

An **entry-controlled loop** checks the condition before executing the loop body. If the condition is false initially, the loop body is not executed.

```
#include <stdio.h>

int main()
{
    int i = 5;

    while(i < 5)
    {
        printf("\n%d\n ", i);
        i++;
    }

    return 0;
}
```

OUTPUT: NO OUTPUT, because the condition is false at the start.

```
PS E:\OOPJ_Net_Beans_practical-project\TOPS_workshop\ASSIGNMENTS\Software_engineering\Module_2> cd "e:\OOPJ_Net_Beans_practical-project\TOPS_workshop\ASSIGNMENTS\Software_enginee
ring\Module_2\" ; if ($?) { gcc Assignment6_Tasks.c -o Assignment6_Tasks } ; if ($?) { .\Assignment6_Tasks }
PS E:\OOPJ_Net_Beans_practical-project\TOPS_workshop\ASSIGNMENTS\Software_engineering\Module_2> 
```

2. exit-controlled loop (do-while)

An **exit-controlled loop** executes the loop body first and checks the condition afterward. Therefore, the loop body always executes at least once.

```
#include <stdio.h>

int main()
{
    int i = 5;

    do
    {
        printf("\n%d\n ", i);
        i++;
    } while(i < 5);

    return 0;
}
```

OUTPUT: Because the condition is checked after the first execution, the loop runs once even though the condition is false initially.

```
5
```